WARNING: Sorry, the genes arnD, arnE, fliR, gtrB, hcp, iagB, kdgM, kdgN, lptC, lptF, lptG, lpxA, mrdB, nanC, phoQ, prgI, rfaF, rfaQ, rfaY, rfbG, rfbN, rfbV, sigE, siiC, siiD, siiF, sopF, spaN, spiA, ssaD, ssaE, ssaI, ssaK, ssaL, ssaT, ssaU, ssas, sseE, sseK1, sseL, tagH, tisB, wbaV aren't in our dataset, which is focused on lead-operon promoters.
Please check out the other [SalCom databases] for our other comprehensive transcriptomic data.

Reference:


nature microbiology

Resource

Profiling *Salmonella* transcriptional dynamics during macrophage infection using a comprehensive reporter library

[PDF (15 MB)]

Welcome to SalComKinetics

SalcomKinetics is an interactive browser for the gene expression data presented in [Temporal profiling of Salmonella transcriptional dynamics during macrophage infection using a comprehensive reporter library].
These data are collected from our GFP-reporter assay from 2,907 computationally identified promoter regions.

Instructions: Search by gene, locus or location (multiple entries separated by comma or space, use '*' for wild-card).

*To start: enter query or select Example: Top 100 genes in the early stage (1-4 h.p.i) • Top 100 genes in the middle stage (5-9 h.p.i) • Top 100 genes in the late stage (10-12 h.p.i) • Top 100 genes in the escape stage (13-15 h.p.i) • Enter-Doudoroff • Mn2+ regulation • SPI1 genes • SPI2 • SPI1 & 2 effectors • SPI3 • SPI5 • SPI6 • SPI11

arcB, arnD, arnE, bssR, csgD, fliR, gtrB, gtrB, hcp, iagB, invF, invH, kdgM, kdgN, nanC, ompL, lptC, lptF, lptG, lpxA, mgrB, n

Query

Heat map for expression of 32 genes in SL1344

Macrophage Expression Levels

	query	01h	02h	03h	04h	05h	06h	07h	08h	09h	10h	11h	12h	13h	14h	15h	Gene_Plate	Gene	Synonym	Ty2 Accession	LT2 Accession	Paratypl
1	arcB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	arcB_P14	arcB		T_RS16455	STM3328	SEEPA5
2	bssR	-	22.1	6.9	7.5	6.7	5.5	6	6.9	6.4	7.8	5.7	5.6	6.3	4.6	6.4	bssR_P27	bssR		bssR	STM0853	bssR
3	csgD	-	-	-	-	-	-	-	-	1.1	1.6	1.7	2.7	2.9	2.5	2.6	csgD_P11	csgD		T_RS09045	STM1142	SEEPA5
4	invF	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	invF_P12	invF		T_RS14210	STM2899	SEEPA5
5	invH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	invH_P12	invH		T_RS14215	STM2900	SEEPA5
6	ompL	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	ompL_P22	ompL		ompL	STM4016	ompL
7	mgrB	-	-	-	-	2.4	5	10	15.9	17.6	12.9	11.1	7.1	7.2	6.9	4.6	yobG_P8	yobG	mgrB	T_RS05300	STM1840	SEEPA5
8	mrcA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	mrcA_P4	mrcA		T_RS20405	STM3493	SEEPA5
9	ompR	-	-	-	-	-	-	-	1.1	-	-	0.7	0.6	-	0.6	-	ompR_P15	ompR		T_RS20360	STM3502	SEEPA5
10	ompX	-	-	-	-	4	6.6	11	14.7	16	11.3	-	12.5	11.9	11	11	ompX_P21	ompX		ompX	STM0833	ompX
11	pagP	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	pagP_P27	pagP				
12	phoP	-	-	-	-	-	15.2	31.7	45.1	78.8	88.3	77	77	74.3	63.5	55.6	phoP_P6	phoP		T_RS08600	STM1231	SEEPA5
13	pipA	-	-	-	-	1.8	-	-	-	-	-	-	-	-	-	-	pipA_P27	pipA		T_RS09330	STM1087	SEEPA5
14	pipB	-	-	-	-	3.7	8.6	13.9	26.6	39.9	65.9	78.5	101.6	103.6	104.7	92.6	pipB2_P12	pipB2		T_RS13590	STM2780	SEEPA5
15	pipB	-	-	-	-	-	7.6	13.8	20.7	20.6	16.8	16.1	12.4	13.4	12	12.3	pipB_P7	pipB		T_RS09325	STM1088	SEEPA5
16	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	rbsK_P16	rbsK		T_RS18950	STM3793	SEEPA5
17	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	EMPTY_P16	EMPTY		NaN	NaN	NaN
18	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	rbsK_P4	rbsK		T_RS18480	STM3885	SEEPA5
19	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	orf408_P31	orf408	rbsK	T_RS06865	STM1382	SEEPA5
20	rfbU	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	rfbU_P31	rfbU		T_RS03965	STM2086	SEEPA5
21	rfc	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	rfc_P5	rfc		T_RS06195	STM1332	SEEPA5
22	siiA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	siiA_P17	siiA		T_RS21165	STM4257	SEEPA5
23	sipD	11.6	12.6	19.2	9.3	5.8	6.6	4.6	4.4	3.8	3.5	2.1	1.9	1.5	1.2	1.2	sipD_P25	sipD		T_RS14130	STM2883	SEEPA5
24	sopD2	7.7	5.9	-	-	2.7	4.2	5.3	6.7	10	10.7	11	7.6	7	7.6	7.7	sopD2_P6	sopD2		T_RS09990	STM0972	SEEPA5
25	sopE	-	-	-	-	-	1.8	1.6	2	-	-	-	-	-	-	-	sopE_P13	sopE		T_RS21905		SEEPA5
26	sopE2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	sopE2_P8	sopE2		T_RS09265	STM1855	SEEPA5
27	spaM	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	P30G1_P30	P30G1		spaM		SEEPA5
28	ssaG	-	-	-	-	10.2	19.2	35.8	47.9	40.5	40.6	35.8	33.3	29.9	31.4	29.7	ssaG_P8	ssaG		T_RS06505	STM1406	SEEPA5
29	ssaM	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	ssaM_P2	ssaM		T_RS06540	STM1413	SEEPA5

≤ 0 10 ≥ 100

Promoter activity dynamics throughout intracellular macrophage infection. During the early stage of infection (1-4 h.p.i.), activity was quantified as the background-subtracted GFP signal. Promoter activity during the middle, late, and escape stages (5-15 h.p.i.) was quantified as the background-subtracted GFP signal, normalized by background-subtracted dTomato signal to account for intracellular replication. Gray boxes indicate that the signal is "OFF," meaning the promoter activity is below the dynamic background threshold. For more details about normalization, please see Methods [here].

	query	01h	02h	03h	04h	05h	06h	07h	08h	09h	10h	11h	12h	13h	14h	15h	Gene_Plate	Gene	Synonym	Ty2 Accession	LT2 Accession	Paratypl
30	*ssaR*	-	-	-	4.1	15.1	24.1	37.4	45.4	36.4	38.3	34.4	28.7	33.1	26.8	26.5	ssaR_P2	ssaR		T_RS06570	STM1419	SEEPA5
31	*sscB*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	sscB_P22	sscB		T_RS06490	STM1403	SEEPA5
32	*sseJ*	-	-	-	-	10.1	22.7	33.4	51	54.8	49.7	38.2	44	42.8	42.8	34.2	sseJ_P8	sseJ		T_RS08125	STM1631	SEEPA5
33	*steB*	-	5.5	-	-	-	-	-	-	-	-	-	-	-	-	-	steB_P6	steB			STM1629	
34	*tagK*	4.2	9.8	10.2	8	1.8	-	-	-	-	-	-	-	-	-	-	sciF_P21	sciF		T_RS13180	STM0271	SEEPA5
35	*waaB*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	rfaB_P20	rfaB		T_RS19330	STM3719	SEEPA5

≤ 0 10 ≥ 100

Promoter activity dynamics throughout intracellular macrophage infection. During the early stage of infection (1-4 h.p.i.), activity was quantified as the background-subtracted GFP signal. Promoter activity during the middle, late, and escape stages (5-15 h.p.i.) was quantified as the background-subtracted GFP signal, normalized by background-subtracted dTomato signal to account for intracellular replication. Gray boxes indicate that the signal is "OFF," meaning the promoter activity is below the dynamic background threshold. For more details about normalization, please see Methods here.

d

***In Vitro* Expression Levels**

	query	00h InSPI2	01h InSPI2	02h InSPI2	03h InSPI2	04h InSPI2	05h InSPI2	06h InSPI2	07h InSPI2	08h InSPI2	09h InSPI2	10h InSPI2	11h InSPI2	12h InSPI2	13h InSPI2	14h InSPI2	15h InSPI2	16h InSPI2	17h InSPI2	18h InSPI2	19h InSPI2	20h InSPI2	21h InSPI2	22h InSPI2	23h InSPI2	24h In	
1	*arcB*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2	*bssR*	2,787	2,323	4,103	6,290	9,833	16,923	22,114	27,513	29,933	31,552	28,428	22,943	18,428	14,629	12,000	11,800	11,600	11,400	10,800	10,800	10,900	10,900	10,200	10,500	9,780	1,99
3	*csgD*	30	73	180	673	1,213	2,467	4,753	6,530	7,676	7,129	8,029	7,661	6,751	5,153	4,110	4,060	4,140	3,690	3,770	3,940	3,550	3,660	3,550	3,620	3,620	4
4	*invF*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
5	*invH*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
6	*ompL*	600	290	547	653	373	710	767	539	251	351	86	542	384	349	407	170	307	152	152	220	240	282	542	543	525	-
7	*mgrB*	207	300	287	640	937	1,393	1,887	2,026	1,888	1,866	1,733	1,975	1,679	1,123	1,200	781	510	859	829	749	704	791	848	805	889	10
8	*mrcA*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
9	*ompR*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	10
10	*ompX*	327	200	513	413	1,420	3,850	7,790	13,980	15,086	16,067	16,227	17,139	15,920	11,445	9,270	7,780	7,700	7,290	7,590	7,680	7,720	7,810	7,960	8,000	8,350	-11
11	*pagP*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
12	*phoP*	23	-250	-130	253	1,760	3,033	5,823	7,559	7,519	6,940	7,505	7,198	5,936	4,326	3,560	3,240	2,960	3,100	3,200	3,290	3,260	2,890	3,140	2,900	2,880	31
13	*pipA*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
14	*pipB*	-310	-623	-670	-563	-667	-490	207	1,266	2,300	2,861	3,371	3,386	2,959	2,302	1,940	1,440	1,140	956	1,090	1,040	1,250	978	1,190	1,070	1,100	11
15	*pipB*	-247	-187	-103	-250	173	497	1,370	2,274	2,996	3,434	3,395	2,813	2,161	1,377	1,120	788	856	1,110	906	915	988	951	1,030	762	774	-28
16	*rbsK*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
17	*rbsK*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
18	*rbsK*	450	413	290	650	737	1,023	935	655	656	740	544	404	234	339	438	394	470	398	346	366	207	244	497	240	332	-
19	*rbsK*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20	*rfbU*	307	390	197	563	710	747	614	348	292	255	126	-173	76	-48	-94	-20	-22	29	-78	0	-124	-184	-105	-65	-183	-
21	*rfc*	370	293	110	297	247	33	577	499	805	765	693	744	901	826	590	264	594	657	230	281	367	529	370	330	150	-
22	*siiA*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
23	*sipD*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
24	*sopD2*	-630	-767	-690	-783	-620	-567	497	997	2,272	3,022	3,945	3,426	3,065	2,390	1,960	1,690	1,460	1,560	1,730	1,490	1,790	1,550	1,850	1,440	1,480	16
25	*sopE*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	92
26	*sopE2*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
27	*spaM*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
28	*ssaG*	190	187	373	460	1,097	2,050	3,785	5,971	7,562	8,849	8,553	7,864	5,844	4,452	3,330	3,100	2,900	3,070	2,890	2,790	2,870	2,710	2,400	2,790	2,580	44
29	*ssaM*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
30	*ssaR*	-97	-230	43	107	560	1,577	4,507	5,774	6,967	7,815	7,464	6,477	5,264	4,132	3,220	3,220	3,090	3,050	3,300	3,250	3,150	3,470	3,200	3,000	3,170	-17

≤ 0 10k ≥ 100k

Promoter activity dynamics throughout growth in in vitro conditions (InSPI2, InSPI2 low Mg, NonSPI2, and 10% BHI). Activity was quantified as t background-subtracted GFP signal, normalized by background-subtracted OD600 to account for changes in cell density. Gray boxes indicate tha signal is "OFF," meaning the promoter activity is below the dynamic background threshold. For visualization, one datapoint is shown per hour. For the dynamic data, please toggle to full resolution view to visualize all data points, or download the raw data. For more details about normalizatic please see Methods here.

	query	00h InSPI2	01h InSPI2	02h InSPI2	03h InSPI2	04h InSPI2	05h InSPI2	06h InSPI2	07h InSPI2	08h InSPI2	09h InSPI2	10h InSPI2	11h InSPI2	12h InSPI2	13h InSPI2	14h InSPI2	15h InSPI2	16h InSPI2	17h InSPI2	18h InSPI2	19h InSPI2	20h InSPI2	21h InSPI2	22h InSPI2	23h InSPI2	24h In
31	*sscB*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
32	*sseJ*	-140	30	153	147	353	1,203	3,517	6,409	9,000	11,198	11,935	11,903	9,871	7,460	5,790	5,700	5,480	5,250	5,280	5,240	5,220	5,420	5,270	5,290	5,100	-
33	*steB*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
34	*tagK*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
35	*waaB*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

≤ 0 10k ≥ 100k

Promoter activity dynamics throughout growth in in vitro conditions (InSPI2, InSPI2 low Mg, NonSPI2, and 10% BHI). Activity was quantified as t background-subtracted GFP signal, normalized by background-subtracted OD600 to account for changes in cell density. Gray boxes indicate that signal is "OFF," meaning the promoter activity is below the dynamic background threshold. For visualization, one datapoint is shown per hour. For the dynamic data, please toggle to full resolution view to visualize all data points, or download the raw data. For more details about normalizatic please see Methods here.

Results provided by the Huang and Monack Labs, automatically generated on Wed Nov 19 04:21:22 2025